Krishna Subedi

Applied AI Engineer — LLM Systems & Robotics

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Experience

Neryva Jul 2025 – Present

Member of Technical Staff · Remote

- Built a durable Temporal runtime with bounded 13-state kernel, activity-isolated model/tool calls, fail-closed budgets, and idempotent ConnectRPC/SSE streaming, proving exactly-once replay across 10,820 stress and 600 load runs.
- Engineered an agent security perimeter with dual-stage fail-closed moderation, spotlighting against prompt injection, engine-authorized tools with credential scoping, human approvals, and cross-runtime SHA-256 hash-chained audit trail.
- Architected a model gateway for 10 providers (240+ models) with live catalog sync and ordered failover, and hybrid RAG (pgvector, MongoDB Atlas, Qdrant) enforcing pre-scoring ACLs, outbox-driven index sync, and MinHash deduplication

Smaitic Labs Apr 2024 – Jul 2025

Full-stack Developer

- Designed Vanidya's three-tier admin model (main, state, seller) with hierarchical RBAC, tenant data isolation, and tier-specific dashboards, supporting new region and seller onboarding without schema or code changes.
- Built a multi-tenant Spring Boot backend with ZIP-code delivery routing and an automated commission engine replacing manual calculations, shipping real-time order tracking synchronized across native Android (Kotlin) and offline-first PWA buyer/seller views.

Nepal Oil Corporation Limited Oct 2023 – Mar 2024

Software Engineer

- Designed a dynamic time-series forecasting system for regional diesel, petrol, and aviation fuel demand, modeling shock surges from strikes and festivals to optimize depot inventory allocation and prevent consumer-side pump stockouts.
- Engineered automated Python and SQL pipelines consolidating national depot and tanker movements into live operational dashboards, replacing manual end-of-day reporting with continuous dispatch visibility.

Selected Projects & Research

Agent Studio: Enterprise AI Agent Platform Jul 2025 – Present

Work Project & Open Source · ML & Platform

- Architected a multi-tenant LLM control plane implementing CoALA (scoped memory, action spaces, LangGraph loops), vector-isolated RAG, and an LLM gateway featuring provider routing, fallback chains, circuit breakers, and cost tracking.
- Built a session and context engine featuring durable threads, compaction, and conversation forking, integrated with Presidio PII redaction, NeMo Guardrails, human-in-the-loop escalation, and automated Garak and RAGAS evaluation suites.

Krishi Vaidya: AI-Powered Crop Disease Diagnosis Apr – Jul 2026

Personal Project · ML

- Engineered a dual-model crop-disease diagnostic system combining an edge YOLO detector for real-time symptom localization with a cloud Qwen2.5-VL-3B VLM, fine-tuned via LoRA, for structured multimodal assessment.
- Optimized multi-tier inference, benchmarking YOLO across INT4/INT8/FP8 to deploy an INT8 TFLite model for on-device OpenCV execution, serving VLM on vLLM with NF4 quantization, FlashAttention-2, structured JSON, and OOD detection.

Bounded Precision-Geometry Scaling for Robust Multi-Task Learning Aug 2026

Sole Author · arXiv:2608.21653

- Held macro 0.777→0.778 under 1000× loss rescaling, outperforming Nash-MTL on NYUv2.

The Reliability of LLMs for Medical Diagnosis Mar 2025

Sole Author · arXiv:2503.10647

- Evaluated leading LLMs on 52 patient cases; quantified sensitivity to misleading prompts and contextual manipulations.

Education

Bachelor of Engineering in Mechanical Engineering 2019 – 2023

Tribhuvan University · Licensed PE

Skills

Languages: Python, TypeScript/JavaScript, Java, Kotlin, C/C++, SQL · **AI/ML:** PyTorch, Transformers, LLMs/VLMs, LoRA/PEFT, Computer Vision, Time-Series, Multi-Task & Continual Learning · **LLM Systems:** RAG, LangGraph, LLM/Agent Evaluation, Tool Calling, vLLM, Guardrails · **Full-Stack & Web:** React, Spring Boot, FastAPI, Node.js (Express), Redux · **Infrastructure:** Docker, Kubernetes, Redis, PostgreSQL, Git, CI/CD, Linux